

Luigi Roberti

Curriculum Vitæ

Leibniz Universität Hannover
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Personal Information

Place of birth Brescia (BS), Italy
Citizenship Italian
Gender Male
Title Dr.rer.nat.

Current Employment

01/10/2025 – [Institut für Angewandte Mathematik, Leibniz Universität Hannover](#),
Present Germany.
Wissenschaftlicher Mitarbeiter – Postdoc (Postdoctoral Research Assistant).

Past Employment

01/10/2021 – [Faculty of Mathematics, University of Vienna](#), Austria.
31/08/2025 Universitätsassistent Praedoc (Predoctoral University Assistant).
01/10/2020 – [Faculty of Mathematics, University of Vienna](#), Austria.
30/09/2021 Wissenschaftlicher Projektmitarbeiter (Scientific Project Assistant).
Employment within the project “[Equatorial wave-current interactions](#)”,
supported by the WWTF grant MA16-009.
Principal Investigator: Prof. Dr. [ADRIAN CONSTANTIN](#).

Education

10/2020 – **Dr.rer.nat. in Mathematics**, University of Vienna, Austria.
08/2025 Thesis: *Mathematical aspects of geophysical fluid flows*.
Advisor: Prof. Dr. [ADRIAN CONSTANTIN](#).
10/2018 – **MSc in Mathematics**, University of Vienna, Austria.
09/2020 Thesis: *On the decrease of velocity with depth in periodic water waves*.
Advisor: Prof. Dr. [ADRIAN CONSTANTIN](#).
10/2015 – **BSc in Mathematics**, University of Vienna, Austria.
08/2018 Thesis: *Hardy's inequality*.
Advisor: Prof. Dr. [ROLAND DONNINGER](#).
10/2014 – **BSc in Physics**, University of Vienna, Austria.
03/2018 Thesis: *Bestimmung der Volumenverteilung eines Aerosols aus der Messung von dessen Extinktionskoeffizienten*.
Advisor: Prof. Dr. [HELMUTH HORVATH†](#).
09/2009 – **Diploma di Maturità Scientifica (Science High School Diploma)**,
07/2014 Liceo Scientifico Statale Nicolò Copernico, Brescia, Italy.

Publications

Preprints

- [1] (With [C. PUNTINI](#)) Why a mid-depth stress-free boundary condition is incorrect for Ekman flows. [arXiv:2607.29261](#), 6 pp.
- [2] (With [C. PUNTINI](#) and [E. STEFANESCU](#)) On large-scale wind-drift ocean currents: An asymptotic approach in spherical coordinates. [arXiv:2602.06473](#), 42 pp.

Published articles

- [3] (With [Q. DING](#)) Variational formulation for a model describing the Antarctic Circumpolar Current with variable density. *Differ. Integral Equ.* **39**(7/8) (2026), 497–510. [doi:10.57262/die039-0708-497](#).
- [4] (With [E. STEFANESCU](#)) Global-in-time existence, uniqueness, and stability of solutions to a model of the Antarctic Circumpolar Current. *Discrete Contin. Dyn. Syst.* **49** (2026), 289–310. [doi:10.3934/dcds.2025169](#).
- [5] (With [B.-V. MATIOC](#) and [CH. WALKER](#)) Quasilinear parabolic equations with superlinear nonlinearities in critical spaces. *J. Differ. Equ.* **429** (2025), 283–317. [doi:10.1016/j.jde.2025.02.039](#).
- [6] (With [Q. DING](#)) Stratified ocean gyres with Stuart-type vortices. *Ann. Mat. Pura Appl.* **203** (2024), 2847–2862. [doi:10.1007/s10231-024-01469-5](#).
- [7] (With [B.-V. MATIOC](#)) Weak and classical solutions to an asymptotic model for atmospheric flows. *J. Differ. Equ.* **367** (2023), 603–624. [doi:10.1016/j.jde.2023.05.023](#).
- [8] The surface current of Ekman flows with time-dependent eddy viscosity. *Comm. Pure Appl. Anal.* **21**(7) (2022), 2463–2477. [doi:10.3934/cpaa.2022064](#).
- [9] The Ekman spiral for piecewise-constant eddy viscosity. *Appl. Anal.* **101**(15) (2022), 5528–5536. [doi:10.1080/00036811.2021.1896709](#).
- [10] Perturbation analysis for the surface deflection angle of Ekman-type flows with variable eddy viscosity. *J. Math. Fluid Mech.* **23**(3) (2021), No. 57. [doi:10.1007/s00021-021-00586-y](#).
- [11] On the decrease of velocity with depth in irrotational periodic water waves. *Monatsh. Math.* **193**(3) (2020), 671–682. [doi:10.1007/s00605-020-01451-2](#).

Conferences and Seminars

Talks

- 2026 Invited talk at [the 15th AIMS Conference](#), Athens, Greece.
Talk at [SIAM Conference on Nonlinear Waves and Coherent Structures \(NWCS26\)](#), Montréal, Canada.
Invited talk in working seminar, Hangzhou Normal University, China.
Invited talk at workshop Fluid Dynamics and Spectral Theory of Differential Operators, Nanjing University, China.
- 2025 Invited talk at [Bielefeld Analysis Seminar](#), Universität Bielefeld, Germany.

- Talk at [Oberseminar Analysis und Theoretische Physik](#), Leibniz Universität Hannover, Germany.
- Invited talk at workshop [Modelling of Fluid Propagation: Mathematical Theory and Numerical Approximation](#), International Centre for Mathematical Meetings (CIEM), Castro Urdiales, Spain.
- Invited talk at [Conference on Mathematics of Wave Phenomena](#), Karlsruhe Institute of Technology, Germany.
- 2024 Invited talk at [Mathematical Sciences Seminar](#), University College Cork, Ireland.
- Talk at workshop [Mathematical Theory of Water Waves](#), Lund University, Sweden.
- Talk at [PDE Afternoon Seminar](#), University of Vienna, Austria.
- Invited talk at [MCMP Seminar](#), University of Vienna, Austria.
- 2023 Talk at [ÖMG Tagung 2023](#), TU Graz, Austria.
- 2022 Talk at [DMV-Jahrestagung 2022](#), Freie Universität Berlin, Germany.
- Talk at [Seminar Applied and Computational PDE](#), University of Vienna, Austria.
- 2020 Talk at [Seminar: Applied PDE](#), University of Vienna, Austria.
- 2019 Talk at [Complex Analysis Seminar](#), University of Vienna, Austria.

Additional conferences and programmes attended

- 2025 Workshop [Mathematical Advances in Geophysical Fluid Dynamics](#), Mathematisches Forschungsinstitut Oberwolfach, Germany.
- 2024 [EWM-EMS Summer School: Water Waves and Nonlinear Dispersive Equations](#), Mittag-Leffler-Institut, Sweden.
- [Fluid Flows – Analysis and Modelling. Conference in Honour of Robin S. Johnson’s 80th Birthday](#), University of Vienna, Austria.
- 2023 [Aspects of Nonlinear Evolution. Conference in Honour of Joachim Escher’s 60th Birthday](#), Leibniz Universität Hannover, Germany.
- [Workshop on Nonlinear Dispersive Waves](#) (online attendance), University College Cork, Ireland.
- 2021 [Summer School of the Vienna School of Mathematics](#), Weißensee, Austria.

Teaching

- 2026S [Funktionentheorie für das Lehramt](#) (Lecture on complex analysis for prospective teachers, in German), Leibniz Universität Hannover.
- [Übung zu Funktionentheorie für das Lehramt](#) (Exercise class to my lecture on complex analysis for prospective teachers, in German), Leibniz Universität Hannover.
- 2024W [Proseminar “Komplexe und Harmonische Analysis”](#) (Exercise class “Complex and Harmonic Analysis”, in German), University of Vienna.
- 2024S [Proseminar “Analysis 2”](#) (Exercise class “Analysis 2”, in German), University of Vienna.

- 2023W [Proseminar “Analysis und Lineare Algebra 1”](#) (Exercise class “Analysis and Linear Algebra 1”, in German), University of Vienna.
- 2023S [Übungen zu “Funktionalanalysis”](#) (Exercise class to “Functional Analysis”, in German), University of Vienna.
- 2022W [Übungen zu “Partielle Differentialgleichungen”](#) (Exercise class to “Partial Differential Equations”, in German), University of Vienna.
- 2022S [Übungen zur Einführung in das mathematische Arbeiten und Rechenübungen](#) (Exercise class to Introduction to mathematics and calculations, in German), University of Vienna.

--- (Co-)Supervision

- 2025 BSc thesis of Stefano Mazzeo, University of Vienna (co-supervised with Dr. [JÖRG WEBER](#)).

--- Peer Review Activity

Referee for [Appl. Math. Comput.](#) / [Appl. Math. Lett.](#) / [J. Differ. Equ.](#) / [J. Math. Phys.](#) / [Monats. Math.](#) / [Nonlinear Anal. Real World Appl.](#) / [Pure Appl. Geophys.](#) / [Qual. Theory Dyn. Syst.](#)

--- Training

- 2025 Workshop [Diversity in Practice](#), University of Vienna.

--- Computer Skills and Competence

Operating systems	Working knowledge of Windows and Linux OS.
Programming languages	Working knowledge of Python, Fortran, LaTeX language and some of its text editors (Overleaf, Texmaker, TeXworks).
Software programs	Working knowledge of Matlab, Wolfram Mathematica, Gnuplot.

--- Platforms

-  [arXiv](#)
-  [Google Scholar](#)
-  [ORCID](#)
-  [ResearchGate](#)
-  [Scopus](#)

--- Languages

- Italian Native speaker.
- English Excellent knowledge and understanding of both written and spoken language.
- German Excellent knowledge and understanding of both written and spoken language.